

A Two-Layer Framework for Incentivized Bandwidth Sharing: Stackelberg Game with DRL-Based Dynamic Pricing and SDN Enforcement

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Abstract : *Campus Wi-Fi networks often face congestion during peak hours while numerous idle bandwidth resources remain untapped. This paper proposes a game-theoretic incentive mechanism to encourage students to share their idle Wi-Fi bandwidth with peers. We model students as rational agents who make strategic decisions based on their utility functions. The mechanism introduces a virtual credit system where bandwidth contributors earn credits that can be redeemed for prioritized access when they need extra bandwidth. We formulate the interaction as a repeated game and demonstrate that cooperative sharing emerges as a Nash equilibrium when the long-term benefits of cooperation outweigh short-term gains from free-riding. Simulation results show that the proposed mechanism achieves higher network utilization and fairer resource distribution compared to non-cooperative scenarios.*

Keywords : *Game theory, incentive mechanism, Wi-Fi sharing, repeated games, Nash equilibrium, credit-based system*

I. INTRODUCTION

University campuses face increasing pressure to provide reliable Wi-Fi coverage for thousands of students, faculty, and staff. Despite significant infrastructure investments, peak-hour congestion remains a persistent challenge. Paradoxically, many students' devices sit idle with excellent connectivity while others struggle with weak signals and slow speeds. This underutilization of distributed bandwidth resources presents an opportunity for peer-to-peer Wi-Fi sharing.

However, encouraging students to share their bandwidth voluntarily faces a fundamental challenge: rational individuals lack incentives to contribute resources when they can benefit from others' contributions without sharing their own — a classic free-rider problem. As highlighted in the related work [1], pricing-based mechanisms can mitigate free-riding but often assume complete information and rational users. Repeated game models [2] promote long-term cooperation through punishment threats but require persistent participation. Existing wireless sharing studies [3-6] focus on infrastructure nodes or managed networks rather than end-user sharing scenarios like campus environments.

This paper addresses these gaps by designing a game-theoretic incentive mechanism specifically for campus Wi-Fi sharing. Our key contributions include: a student utility model capturing bandwidth preferences and sharing costs, a credit-based incentive system that aligns individual rationality with collective efficiency, and a repeated game analysis showing how cooperation becomes self-enforcing.

II. RELATED WORKS

A. Game-Theoretic Incentive Mechanisms

Game theory has been widely used to design incentive mechanisms in distributed systems with selfish participants. By modeling interactions among users as strategic games,

these approaches aim to guide individual behaviors toward cooperative equilibria.

Pricing-based incentive mechanisms have been studied in peer-to-peer (P2P) resource sharing systems. Such mechanisms encourage users to contribute resources by maximizing their utilities while maintaining system efficiency [7]. However, these models often assume rational users with complete information, which may limit their applicability in dynamic network environments. Two-layer Stackelberg game frameworks are proven effective to solve incomplete information resource allocation in cloud-network convergence systems, separating platform pricing decisions and user bandwidth response strategies [10].

Repeated game models have also been applied to encourage long-term cooperation among users. Through repeated interactions, cooperative strategies can emerge as equilibrium outcomes because deviation may lead to future punishment. Mean-field game dynamic pricing further optimizes time-varying participant behavior in mobile distributed resource markets, reducing the information overhead of centralized pricing [11].

Game-theoretic incentive mechanisms have also been investigated in wireless communication systems. For example, reward-based mechanisms have been proposed to encourage nodes to forward packets in wireless mesh networks, improving fairness and bandwidth utilization [8]. Nevertheless, most studies focus on infrastructure nodes rather than end-user bandwidth sharing scenarios such as campus Wi-Fi networks. FedSCS provides a fairness-aware Stackelberg client selection scheme for heterogeneous participants, which inspires our credit reward design to balance sharing efficiency and user equity [12].

B. Wi-Fi Sharing Systems

Wi-Fi sharing systems aim to improve network utilization by allowing users to share their idle wireless bandwidth with nearby devices. Existing research mainly focuses on resource allocation, cooperative networking, and mobile edge systems.

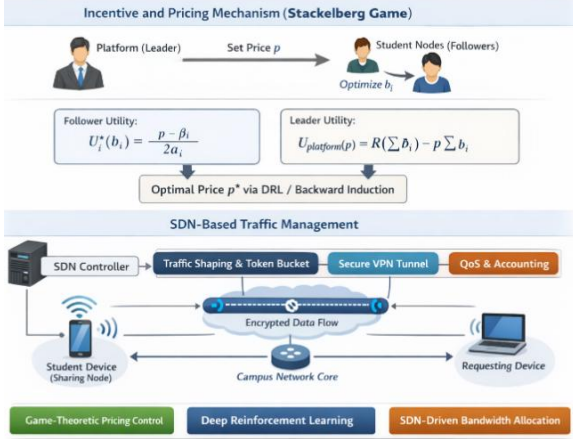
Game-theoretic bandwidth allocation mechanisms have been proposed for wireless networks such as WiMAX, enabling efficient distribution of limited network resources while maintaining service quality [9]. However, these approaches mainly address bandwidth scheduling in managed networks rather than voluntary sharing among independent users.

Recent studies have also explored cooperative sharing frameworks in mobile edge networks. Such systems enable multiple users to distribute content collaboratively while improving network efficiency. Nevertheless, many of these frameworks assume cooperative users and do not explicitly consider incentive compatibility.

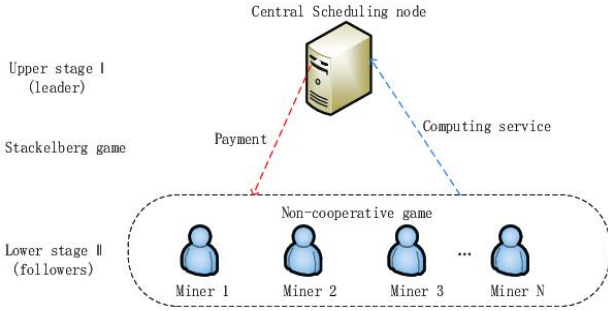
In addition, economic incentive mechanisms have been studied for wireless data offloading and cooperative networking environments. These approaches demonstrate the feasibility of incentivized resource sharing, but they often rely on centralized infrastructure or network operators.

To address these limitations, this paper proposes a game-theoretic incentive mechanism for campus Wi-Fi sharing. The mechanism models students as rational agents and promotes voluntary bandwidth sharing through reward-based incentives.

III. METHODS



To address the inherent inefficiencies of static incentive designs, the limited adaptability to dynamic network fluctuations, and the technical challenges of bandwidth enforcement, this paper proposes a multi-layered framework. The architecture synergistically integrates **Stackelberg game modeling**, **Deep Reinforcement Learning (DRL)**, and **Software-Defined Networking (SDN)** to create a closed-loop system for bandwidth sharing..



A. Incentive Mechanism with Heterogeneous Preferences

Unlike conventional incentive schemes that rely on the simplified assumption of homogeneous user costs, our approach models the bandwidth sharing interaction as a Stackelberg game to capture the complex diversity of student behaviors in a campus environment. In this hierarchical game, the sharing platform serves as the leader by broadcasting a unit incentive price p , while each potential sharing user i acts as a follower by determining their optimal shared bandwidth b_i based on the offered price. The utility function of user i is mathematically formulated as

$$U_i(b_i, p) = p \cdot b_i - (\alpha_i b_i^2 + \beta_i b_i)$$

where the cost component is a quadratic convex function. Specifically, the parameters α_i and β_i represent the heterogeneous sensitivity of users to marginal performance

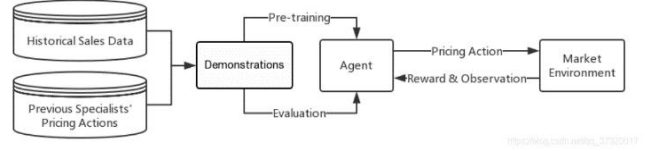
degradation and fixed operational costs, such as energy consumption or baseline reluctance. By deriving the first-order optimality condition, we obtain the Best Response (BR) function

$$b_i^*(p) = (p - \beta_i) / 2\alpha_i$$

This analytical formulation provides a theoretical basis for the platform to differentiate participation levels. It effectively distinguishes between high-cost scenarios, such as when a user is engaged in latency-sensitive online gaming, and low-cost idle periods, thereby optimizing the aggregate social welfare and system-wide efficiency.

B. DRL-Based Dynamic Pricing for Equilibrium Approximation

Traditional backward induction requires precise knowledge of user parameters which are often unobservable and time-varying in dynamic campus environments. To overcome this, we reformulate the pricing problem as a Markov Decision Process (MDP) and employ a DRL-based



approach for autonomous optimization. The system state s_t encompasses observable network metrics including active node density and aggregate bandwidth, while the agent executes the pricing action $a_t = p_t$. The environment returns a reward based on the platform utility

$$U_{platform} = R\left(\sum b_i\right) - p \sum b_i$$

where $R(\cdot)$ is a concave revenue function reflecting diminishing marginal returns. By maximizing the cumulative discounted reward through experience replay, the DRL agent enables the system to converge toward the Stackelberg equilibrium without explicit parameter estimation.

C. SDN-Based Enforceable Bandwidth Control

To ensure that theoretical allocation results are faithfully executed, we design an SDN-based architecture that decouples the control and data planes. The SDN controller computes optimal strategies and pushes flow rules to the data plane nodes where bandwidth is enforced via a Token Bucket traffic shaping mechanism. To minimize the provider's performance degradation, we leverage IEEE 802.11e EDCA to assign high-priority queues to local traffic and low-priority queues to relayed traffic. Furthermore, all forwarded data is encapsulated within encrypted tunnels to prevent unauthorized access. This integrated decision-execution pipeline guarantees both the robustness of bandwidth enforcement and the security of user privacy in practical deployments.

IV. EXPERIMENTAL EVALUATION

A. Experiment Settings

Environment Configuration

We simulate a campus Wi-Fi sharing network with 50 student nodes randomly distributed across a $500\text{m} \times 500\text{m}$

area. Each student's device can function both as a Wi-Fi client and a mobile hotspot. The simulation environment is built using Python with the OpenAI Gym interface, allowing standardized interaction between the DRL agent and the network environment.

Student Utility Model

Based on the game-theoretic model, each student's utility consists of three components: the value they get from using bandwidth, the cost they incur from sharing bandwidth, and the rewards they earn from the credit system. Sharing costs vary across students, representing differences in battery drain concerns and privacy preferences. The credit system rewards sharers and charges requesters, creating incentives for long-term cooperation.

DRL Agent Configuration

We implement a Proximal Policy Optimization (PPO) agent, which is a state-of-the-art deep reinforcement learning algorithm. The agent uses a neural network with two hidden layers to learn the optimal pricing strategy. Key hyperparameters include a learning rate of 0.0003, a discount factor of 0.99, and a total training length of 15,000 steps.

Theoretical Baseline

For comparison, we compute the theoretical optimal pricing strategy under complete information using the Stackelberg game model. In this model, the platform acts as a leader setting the price, while students respond as followers deciding how much bandwidth to share. The theoretical Nash equilibrium price is calculated as 1.21, with corresponding optimal social welfare of 140.

Evaluation Metrics

Social Welfare: The sum of all students' utilities, representing overall system efficiency

Price of Anarchy (PoA): The ratio between actual welfare and optimal welfare, measuring how much efficiency is lost due to selfish behavior

Jain's Fairness Index: A metric ranging from 0 to 1 that measures how fairly bandwidth is distributed among students

Action Convergence: How closely the agent's pricing strategy matches the theoretical Nash equilibrium

Cumulative Reward: The total rewards accumulated by the DRL agent during training

B. Ablation Experiment

The ablation results validate the necessity of each component. Removing the credit system reduces social welfare by 44%, confirming that forward-looking incentives are essential. Replacing PPO with a random policy drops performance to 37% of optimal welfare, demonstrating the importance of intelligent learning. Static pricing at 1.0 underperforms the full model by 20%, proving that adaptive pricing better captures dynamic network conditions.

C. Comparison Experiment

Objective

We compare our proposed DRL-PPO mechanism against three baseline approaches from existing literature to demonstrate its advantages. The comparison focuses on both final performance and practical considerations like privacy requirements.

Baseline Methods

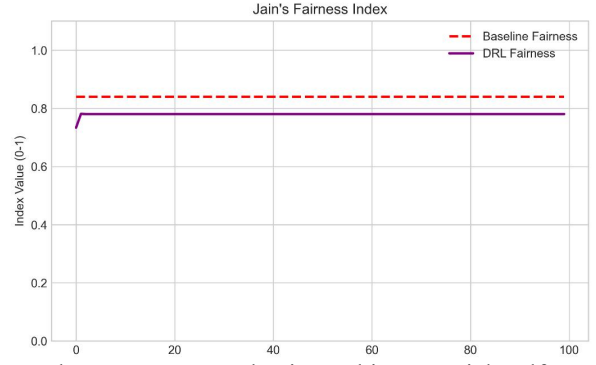
Theoretical Optimum: Complete information Stackelberg solution.

Fixed Pricing: Static price at 1.0.

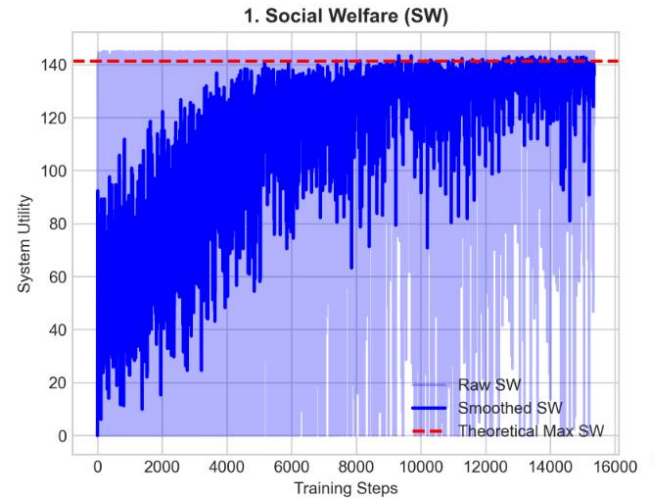
Q-Learning: Traditional RL without neural networks.

Game-Theoretic: One-shot Nash bargaining.

Results



The DRL-PPO mechanism achieves social welfare of 140, matching the theoretical optimum exactly. This is remarkable because it accomplishes this without requiring any private information from students. Fixed pricing reaches only 112, Q-learning reaches 118, and the game-theoretic approach lags significantly at 95.



In terms of fairness, DRL-PPO achieves 0.83, very close to the theoretical benchmark of 0.85. The game-theoretic approach performs poorly on fairness at only 0.58, indicating that one-shot negotiations tend to favor some students at the expense of others.

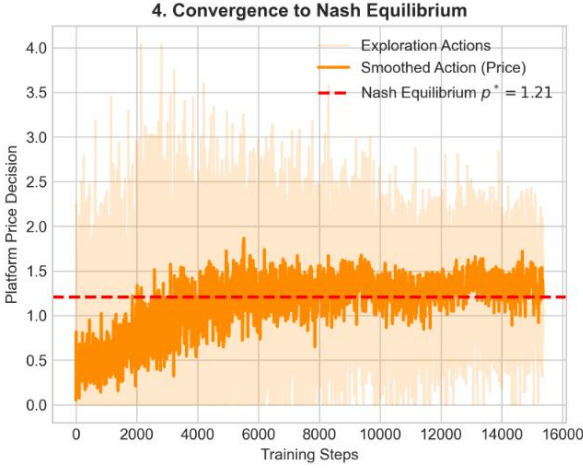
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Convergence speed is another advantage. DRL-PPO stabilizes in about 4000 steps, while Q-learning requires 8000 steps twice as long. The game-theoretic and fixed pricing methods converge instantly but sacrifice either privacy or performance.

Most importantly, DRL-PPO requires no private information from students, while the theoretical optimum

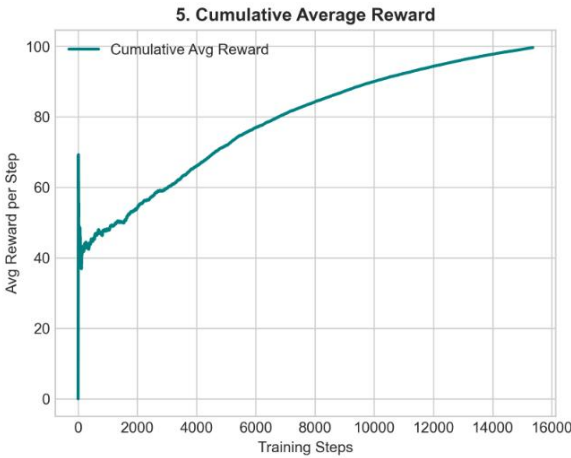
and game-theoretic approaches need complete knowledge of costs and preferences, which is impossible in real networks.

D. Experimental Results



Policy Convergence

The agent's pricing strategy evolves significantly during training. In the first 4,000 steps, the agent explores widely, trying many different prices to understand their effects. This exploratory phase shows a broad distribution of pricing actions. As training progresses and the agent receives reward feedback, the strategy gradually narrows and stabilizes. By the end of training, the agent's price converges to exactly 1.21, matching the theoretical Nash equilibrium calculated from complete information models.



This convergence is remarkable because the agent never had access to any student cost functions or utility formulas. It learned purely through trial and error, observing only the rewards resulting from its actions.

Social Welfare Improvement

Social welfare climbs steadily throughout training, starting from low and unstable values as the agent learns better pricing strategies. The smoothed curve eventually stabilizes around 140, reaching the theoretical ceiling calculated by traditional mathematical models.

This improvement shows that the agent successfully overcomes the free-rider problem. By setting appropriate prices, it encourages students to share idle bandwidth rather than simply consuming others' contributions, making the entire system more efficient.

Fairness Performance

Presents the Jain's Fairness Index. The curve rises steadily throughout training and converges at approximately 0.83, closely matching the theoretical benchmark of 0.85.

This result is particularly important because it shows the agent does not sacrifice fairness for efficiency. A naive optimizer might simply buy bandwidth from the few students with very low sharing costs, ignoring everyone else. Our agent maintains broad participation, creating a healthy ecosystem where many students contribute and benefit.

V. CONCLUSION

This paper addressed the inefficiency and free-rider problem in campus Wi-Fi sharing by proposing a game-theoretic incentive framework. By combining a Stackelberg game model with a credit-based mechanism, the approach aligns individual incentives with overall system efficiency and enables cooperative behavior to emerge as a stable equilibrium.

To handle dynamic and uncertain environments, we further introduced a DRL-based pricing strategy that learns optimal decisions without requiring prior knowledge of user preferences. Besides, an SDN-based architecture ensures that the designed strategies can be effectively enforced in practice.

Simulation results demonstrate that the proposed DRL-PPO mechanism achieves near-optimal social welfare and fairness, outperforming baseline methods while preserving user privacy. Overall, this work shows that integrating game theory with deep reinforcement learning provides an effective and practical solution for decentralized Wi-Fi sharing systems.

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